THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 11 & 12: **Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Power consumed by a 10 Ω resistor carrying 2 A is: 1

(a) 200 W **(b)** 5 W **(c)** 40 W **(d)** 20 W

2. The direction of conventional current is: 1

(a) same as electron flow
(b) from negative to positive terminal outside
(c) opposite to the flow of electrons
(d) random

3. A wire of resistance R is cut into two equal halves, which are joined in parallel. The resistance becomes: 1

(a) R **(b)** $R/4$ **(c)** $2R$ **(d)** $R/2$

4. The potential difference between two points is measured in: 1

(a) ampere **(b)** watt **(c)** volt **(d)** ohm

5. An electric heater rated 1500 W used for 2 h consumes: 1

(a) 1.5 units **(b)** 0.75 units **(c)** 3 units **(d)** 30 units

6. A soft iron bar inside a solenoid becomes a: 1

(a) insulator **(b)** temporary magnet **(c)** permanent magnet **(d)** conductor

7. Which of these produces a uniform magnetic field? 1

(a) long current-carrying solenoid **(b)** bar magnet near poles
(c) single loop at its edge **(d)** straight wire

8. The force on a conductor is reversed when: 1

(a) the direction of current is reversed **(b)** the wire is thicker
(c) the length is doubled **(d)** current is increased

9. Short circuit causes: 1

(a) low voltage **(b)** no effect
(c) a sudden large current **(d)** zero current

10. The tingling sensation when touching a metal appliance indicates: 1

(a) short circuit **(b)** overloading
(c) leakage of current (faulty earthing) **(d)** low voltage

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the equivalent resistance of three 3 Ω resistors in parallel. 2

12. Define 1 ohm. 2

13. Why are copper wires used in solenoids rather than iron wires? 2

14. What is the potential difference between the live and the neutral wire in India? What is the potential of the neutral wire? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Calculate the cost of using five 40 W bulbs for 6 hours a day for 30 days at ₹4 per unit. 3

16. Why does a compass placed near a current-carrying wire deflect? What happens when the current is switched off? 3

17. Why is an ammeter connected in series and a voltmeter in parallel? 3

OR

An electric lamp of 100 Ω, a toaster of 50 Ω and a water filter of 500 Ω are connected in parallel to a 220 V source. Find the resistance of an iron connected to the same source that takes as much current as all three. 3

18. Name the devices that use electromagnets (three). 3

OR

A 15 A circuit is used for heaters while a 5 A circuit is used for bulbs. Explain why. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Why do the filaments of bulbs glow? 5
(ii) Why is an inert gas filled in bulbs?
(iii) Find the heat developed in 5 minutes by a bulb of 220 V, 60 W.

OR

(i) What is meant by potential difference? 5
(ii) How much energy is given to each coulomb of charge passing through a 6 V battery?
(iii) Calculate the number of electrons in 1 C.

20. (i) Explain overloading and short circuit. 5
(ii) How does a fuse protect against them?
(iii) Find the fuse rating for an appliance of 1 kW at 220 V.

OR

(i) What is the magnetic effect of current? 5
(ii) Describe the pattern of the field due to a current in a circular loop.
(iii) Give two applications of the magnetic effect of current.

END OF QUESTION PAPER — Total: 40 Marks